

Lem Convergencia Uniforme

Compactos Subsubsucesiones

Lema 1. Sea $(f_n)_{n \in \mathbb{N}} \subset \mathcal{C}(\Omega)$ y $f \in \mathcal{C}(\Omega)$, entonces

$$f_n \xrightarrow[n \rightarrow \infty]{\Omega} f \iff \forall (n_k)_{k \in \mathbb{N}} \subset \mathbb{N} \text{ creciente} : \exists (n_{k_j})_{j \in \mathbb{N}} \subset (n_k)_{k \in \mathbb{N}} : f_{n_{k_j}} \xrightarrow[j \rightarrow \infty]{\Omega} f.$$

Demostración: La implicación $\Rightarrow$ es trivial. Para la implicación $\Leftarrow$, supongamos por contradicción que f_n no converge a f uniformemente sobre compactos, entonces

$$\exists K \Subset \Omega, \varepsilon > 0 : \forall k \in \mathbb{N} : \exists z_k \in K : \exists n_k \geq N : |f_{n_k}(z_k) - f(z_k)| \geq \varepsilon.$$

Es decir, tomando dichos $K \Subset \Omega$ y $\varepsilon > 0$, existe una sucesión $(n_k)_{k \in \mathbb{N}} \subset \mathbb{N}$ creciente tal que $\forall k \in \mathbb{N} : \exists z_k \in K : |f_{n_k}(z_k) - f(z_k)| \geq \varepsilon$. Entonces, como K es compacto, $\exists a \in K, (z_{k_j})_{j \in \mathbb{N}} \subset (z_k)_{k \in \mathbb{N}} : z_{k_j} \xrightarrow[j \rightarrow \infty]{} a$. Además, por hipótesis,

$$\exists (n_{k_{j_\ell}})_{\ell \in \mathbb{N}} \subset (n_{k_j})_{j \in \mathbb{N}} : f_{n_{k_{j_\ell}}} \xrightarrow[\ell \rightarrow \infty]{\Omega} f.$$

Por el **lem-convergencia-uniforme-compactos-imp-continuidad/??**, $f(z_{k_{j_\ell}}) \xrightarrow[\ell \rightarrow \infty]{} f(a)$

y por el **lem-convergencia-uniforme-compactos-imp-convergencia-sucesion/??**, $f_{n_{k_{j_\ell}}}(z_{k_{j_\ell}}) \xrightarrow[\ell \rightarrow \infty]{} f(a)$. Entonces, debe existir $\ell_0 \in \mathbb{N}$ tal que

$$\forall \ell \geq \ell_0 : \left| f_{n_{k_{j_\ell}}}(z_{k_{j_\ell}}) - f(z_{k_{j_\ell}}) \right| \leq \left| f_{n_{k_{j_\ell}}}(z_{k_{j_\ell}}) - f(a) \right| + \left| f(a) - f(z_{k_{j_\ell}}) \right| < \varepsilon,$$

lo que es una contradicción. ■

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.